

The long lifespan and low turnover of human islet beta cells estimated by mathematical modelling of lipofuscin accumulation.

AIMS/HYPOTHESIS: Defects in pancreatic beta cell turnover are implicated in the pathogenesis of type 2 diabetes by genetic markers for diabetes. Decreased beta cell neogenesis could contribute to diabetes. The longevity and turnover of human beta cells is unknown; in rodents <1 year old, a half-life of 30 days is estimated. Intracellular lipofuscin body (LB) accumulation is a hallmark of ageing in neurons. To estimate the lifespan of human beta cells, we measured beta cell LB accumulation in individuals aged 1-81 years. **METHODS:** LB content was determined by electron microscopical morphometry in sections of beta cells from human (non-diabetic, n = 45; type 2 diabetic, n = 10) and non-human primates (n = 10; 5-30 years) and from 15 mice aged 10-99 weeks. Total cellular LB content was estimated by three-dimensional (3D) mathematical modelling. **RESULTS:** LB area proportion was significantly correlated with age in human and non-human primates. The proportion of human LB-positive beta cells was significantly related to age, with no apparent differences in type 2 diabetes or obesity. LB content was low in human insulinomas (n = 5) and alpha cells and in mouse beta cells (LB content in mouse <10% human). Using 3D electron microscopy and 3D mathematical modelling, the LB-positive human beta cells (representing aged cells) increased from >or=90% (<10 years) to >or=97% (>20 years) and remained constant thereafter.

CONCLUSIONS/INTERPRETATION: Human beta cells, unlike those of young rodents, are long-lived. LB proportions in type 2 diabetes and obesity suggest that little adaptive change occurs in the adult human beta cell population, which is largely established by age 20 years.